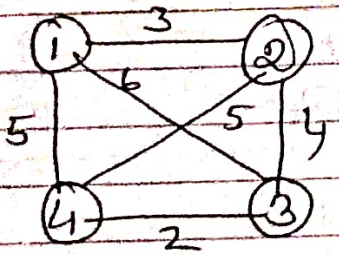
Q What is Exhaustive Search approach to solve the travelling Salesman problem with an Example ? what is efficiency Class of this approach.

Sol) The brute force method can be used to solve Combinational problems called Exhaustive Search.

Travelling Salesman problem (TSP)

It can be defined as "Given n cities, a salesperson start from specified called source and has to visit all the cities and comeback to same city. The objective of this problem is to find a route through cities that minimize the cost.

eg Consider the graph given



So The graph can be represent by (graph) Cost adjacency matrix

	1	2	3	4
1	0	3	6	5
2	3	0	4	5
3	6	4	0	2
4	5	5	2	0

If Salesperson start from city 1, the various routes which can visit all cities without repeating same city and returns back to start city 1 are

$1 \xrightarrow{3} 2 \xrightarrow{4} 3 \xrightarrow{2} 4 \xrightarrow{5} 1 = 14$

$1 \xrightarrow{3} 2 \xrightarrow{5} 4 \xrightarrow{2} 3 \xrightarrow{6} 1 = 16$

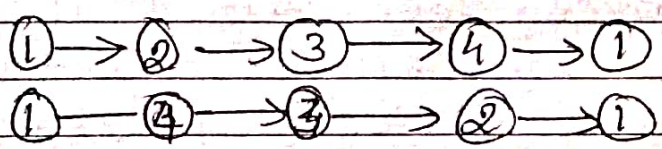
$1 \xrightarrow{6} 3 \xrightarrow{4} 2 \xrightarrow{5} 4 \xrightarrow{5} 1 = 20$

$1 \xrightarrow{6} 3 \xrightarrow{2} 4 \xrightarrow{5} 2 \xrightarrow{3} 1 = 16$

$1 \xrightarrow{5} 4 \xrightarrow{5} 2 \xrightarrow{4} 3 \xrightarrow{6} 1 = 20$

$1 \xrightarrow{5} 4 \xrightarrow{2} 3 \xrightarrow{4} 2 \xrightarrow{3} 1 = 14$

Now, we have to find out smallest among all costs it is 14 and two different path is



Travelling Salesmen can select any one among two

Time efficiency: in any complete graph the no of paths from source to destination (any) other nodes and back to source node is $(n-1)$

for Example if $n=4$ n is no of vertices then different paths are 6

$\therefore$ Complexity is given by $T(n) = O(n-1)!$

$T(n) = O(n!)$